CO1-AT1

CSA0509-Database Management

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Question 1: Explain Schema and Instance with Examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

Schema

A Schema is the overall structure (blueprint) of a database. It defines how the database is organized, including the tables, columns (attributes), relationships, constraints, and data types.

A schema is designed before data is entered into the database and changes very rarely.

Example

Suppose a college stores student information.

Student Table Schema

Student_ID	Name	Department	Age
Integer	Varchar	Varchar	Integer

This defines the structure of the Student table.

Characteristics of Schema

- Blueprint of the database.
- Created by the Database Administrator (DBA).
- Changes very rarely.
- Defines tables, attributes, keys, and relationships.

Instance

An Instance is the actual data stored in the database at a particular point in time.

Whenever records are inserted, updated, or deleted, the instance changes.

Example

Student Table Instance

Student_ID	Name	Department	Age
101	Ravi	CSE	20
102	Priya	ECE	21
103	Arun	IT	19

Tomorrow, if another student joins,

Student_ID	Name	Department	Age
104	Kiran	CSE	20

The instance changes, but the schema remains the same.

Difference Between Schema and Instance

Schema	Instance
Blueprint of database	Actual data stored
Permanent structure	Temporary data
Changes rarely	Changes frequently
Created during database design	Changes during database operations

Schema	Instance
Designed by DBA	Updated by users/applications

Types of Schema

DBMS follows the Three-Schema Architecture.

```
        Users
          |
     View Schema
          |
    Logical Schema
          |
   Physical Schema
          |
     Storage Disk
```

1. Logical Schema (Conceptual Schema)

The Logical Schema describes what data is stored and the relationships among the data.

It does not specify where the data is stored.

Example

STUDENT

Student_ID

Name

Department

Age

COURSE

Course_ID

Course_Name

Relationship

Student ---- Enrolls ---- Course

Characteristics

- Defines tables
- Defines attributes
- Defines relationships
- Independent of hardware

2. Physical Schema

The Physical Schema describes how the data is actually stored inside the storage devices.

It specifies

- Files
- Blocks
- Indexes
- Memory allocation
- Disk organization

Example

Hard Disk

Student.dat

Page 1

Page 2

Page 3

Index File

Student_ID → Page Number

101 → Page1

102 → Page2

Characteristics

- Hardware dependent
- Managed by DBA
- Improves performance

3. View Schema (External Schema)

The View Schema provides only the required information to different users.

Different users can have different views.

Example

Teacher View

Student_ID

Name

Marks

Account Section View

Student_ID

Name

Fees

Students cannot see salary details of teachers.

Characteristics

- Provides security
- Hides unnecessary information
- Different users get different views

Comparison of Schemas

Logical Schema	Physical Schema	View Schema
Defines database structure	Defines storage method	Defines user-specific view
Hardware independent	Hardware dependent	User dependent
Contains tables and relationships	Contains files and indexes	Contains selected data only
Used by designers	Used by DBA	Used by end users

Question 2: Identify the Entities, Attributes, and Relationships for a Bank Management System and Construct its ER Diagram.

What is an ER Diagram?

An Entity Relationship (ER) Diagram is a graphical representation of entities, attributes, and relationships in a database.

Step 1: Identify Entities

Customer

Attributes

- Customer_ID (Primary Key)
- Name
- Address
- Phone

Account

Attributes

- Account_Number (Primary Key)
- Account_Type
- Balance

Loan

Attributes

- Loan_ID
- Amount
- Interest

Employee

Attributes

- Employee_ID
- Name
- Designation

Branch

Attributes

- Branch_ID
- Branch_Name
- Location

Step 2: Identify Relationships

Customer owns Account

One customer may own many accounts.

Customer (1) -------- (M) Account

Customer takes Loan

Customer (1) ------- (M) Loan

Branch maintains Account

Branch (1) -------- (M) Account

Employee works in Branch

Employee (M) ------ (1) Branch

Complete ER Diagram

```
                CUSTOMER

            Customer_ID (PK)

            Name

            Address

            Phone

               |

              Owns

               1:M

               |

               |
```

ACCOUNT

Account_No (PK)

Account_Type

Balance

|

Maintained By

|

|

BRANCH

Branch_ID (PK)

Branch_Name

Location

^

|

Works In

|

EMPLOYEE

Employee_ID (PK)

Name

Designation

CUSTOMER

|

Takes

|

LOAN

Loan_ID

Amount

Interest

Question 3: Explain the Extended ER (EER) Model. Describe Generalization, Specialization, and Aggregation.

Extended ER (EER) Model

The Extended Entity Relationship (EER) Model is an extension of the ER model that includes advanced concepts such as inheritance, subclasses, superclasses, specialization, generalization, aggregation, and categories.

It is mainly used to model complex real-world databases.

Features of EER

- Inheritance
- Generalization
- Specialization
- Aggregation
- Categories
- Constraints

Generalization

Generalization combines multiple lower-level entities into a higher-level entity based on common features.

Example

Savings Account

ACCOUNT

Current Account

Common attributes

- Account Number
- Balance

Specialization

Specialization divides one entity into multiple specialized entities.

Example

```
        EMPLOYEE
             |
     ----------------------
  |                         |
 MANAGER           CLERK
```

Manager

- Bonus
- Team Size

Clerk

- Shift
- Counter Number

Aggregation

Aggregation treats a relationship as an entity.

Example

Employee

|

Works On

|

Project

Aggregation

Assignment

Employee ------ Works On ------ Project

___________________________/

Assignment

This Assignment can further participate in another relationship.

Inheritance

Child entities inherit all attributes of the parent entity.

Example

PERSON

Name

Age

Address

|

| |

Student Teacher

Student inherits

- Name
- Age
- Address

Teacher also inherits

- Name
- Age
- Address

Question 4: Design an EER Diagram for Hospital Management System Incorporating Inheritance and Weak Entities.

Superclass

PERSON

Attributes

- Person_ID
- Name
- Age
- Gender

Subclasses

PATIENT

- Disease
- Admission Date

DOCTOR

- Specialization
- Experience

ROOM

- Room Number
- Room Type

MEDICAL RECORD

Weak Entity

Attributes

- Record Number
- Diagnosis
- Prescription

Medical Record cannot exist without Patient.

EER Diagram

```
                PERSON
          Person_ID (PK)
          Name
          Age
          Gender
               |
       ----------------------
      |                       |
   PATIENT            DOCTOR
 ----------------     ----------------
 Disease              Specialization
 Admission_Date       Experience
     |
     | Has
```

|

MEDICAL_RECORD

(Weak Entity)

Record_No (Partial Key)

Diagnosis

Prescription

PATIENT

|

Admitted To

|

ROOM

Room_No

Room_Type

Explanation

- Person is the superclass.
- Patient and Doctor inherit all attributes of Person.
- Medical_Record is a weak entity because it depends on Patient.
- Room stores patient admission details.

Question 5: Explain the Role of a DBA (Database Administrator) and the Responsibilities Associated with the Position.

Database Administrator (DBA)

A Database Administrator (DBA) is a person responsible for installing, maintaining, securing, monitoring, and managing the database system so that data remains safe, accurate, and available.

The DBA ensures that the database performs efficiently and supports multiple users without data loss or security issues.

Responsibilities of a DBA

1. Database Installation

- Installs DBMS software (such as MySQL, Oracle, SQL Server).
- Configures the database server.

2. Database Design

- Creates tables, relationships, indexes, and constraints.
- Designs the database schema.

3. Security Management

- Creates user accounts.
- Grants and revokes permissions.
- Protects data from unauthorized access.

4. Backup and Recovery

- Takes regular backups.
- Restores the database after system failures.

5. Performance Tuning

- Optimizes SQL queries.
- Creates indexes.
- Monitors database performance.

6. Storage Management

- Allocates disk space.
- Monitors storage usage.
- Manages files and memory.

7. Data Integrity

- Maintains data consistency.
- Enforces constraints (Primary Key, Foreign Key, Unique, Check).

8. Monitoring and Maintenance

- Monitors database health.
- Fixes errors and failures.
- Applies software updates and patches.

9. Disaster Recovery

- Plans for recovery after hardware failures, power outages, or cyberattacks.
- Ensures minimal downtime.

10. User Support

- Assists users with database-related issues.
- Provides access and resolves permission problems.

Diagram: Role of a DBA

```
              DATABASE ADMINISTRATOR (DBA)
                             |
   ------   --------    --------  ---------  -----------    ---------
   |             |           |           |               |        |
 Installation  Security  Backup  Performance Storage  Recovery
   |          |       |             |           |          |      |
 Install DBMS  Manage   Backup  Optimize  Allocate  Restore
            Users     Data    Queries    Space    Data
```

Advantages of a DBA

- Ensures data security and privacy.
- Maintains high database performance.

- Prevents data loss through backups.
- Ensures data consistency and integrity.
- Supports reliable multi-user access.
- Reduces system downtime through proper maintenance.

Conclusion

A **Database Management System (DBMS)** is an essential software system used to store, organize, manage, and retrieve data efficiently. Understanding concepts such as **schema, instance, logical schema, physical schema, and view schema** helps in designing databases that are structured, secure, and easy to maintain. The **Entity-Relationship (ER) model** and **Extended ER (EER) model** provide effective ways to represent real-world entities, relationships, inheritance, and weak entities, making database design more accurate and scalable. The **Database Administrator (DBA)** plays a vital role in ensuring database security, performance, backup, recovery, and overall system reliability. Together, these concepts form the foundation of modern database systems, enabling organizations to manage large volumes of data efficiently while ensuring integrity, consistency, and security.